

??
 Does
 GRPO
 help
 every
 language
 equally?
 Does
 the
 model
 predict
 in
 English
 in
 the
 middle
 layers,
 and
 does
 GRPO
 change
 this?
 If
 a
 language
 does
 not
 improve,
 what
 is
 happening
 inside
 the
 model?

0.46
 0.80
 0.02
 620
 620
 ??

??
 ??
 ??
 ??
 ??
 ??
 ??

π_{θ}
 $\pi_{\theta_{old}}$
 π_{ref}

i
 G
 $\{y_{i,j}\}_{j=1}^G$
 R_i^j

$$\mu^i = \frac{1}{G} \sum_{j=1}^G R_i^j, \sigma^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_i^j - \mu^i)^2}, A_i^j = \frac{R_i^j - \mu^i}{\sigma^i}.$$

(1)

$$\mathcal{L}_{GRPO}(\theta) = -\frac{1}{B} \sum_{i=1}^B \frac{1}{G} \sum_{j=1}^G \mathcal{L}_{i,j}^{resp},$$

(2)

$\frac{B}{0.2}$
 $\beta = 0.04$
 ??
 ??
 1,400
 350
 Step-by-step
 Answer:
 Question:
 Step-by-step
 Answer:
 Final
 Answer:
 Final
 Answer:
 <number>

??
 2,500
 46
 14

LanguageCoverage classScript tokens (approx.)

~ 25,000
 ~ 6,000
 ~ 4,000
 ~ 2,500
 ~ 46
 ~ 14

$L = 28$
 $d = 3584$
 $64 = 64$
 16
 0.05
 q_proj